Short | Box-Cox gamma sweep | Test ANN10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12/18
1 ANN10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
1 ANN10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
1 ANN10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
1 ANN10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
1 ANN10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
1 ANN10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
1 ANN10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
1 ANN10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
1 ANN10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
2 P-only DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13/18
2 P-only DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
2 P-only DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
2 P-only DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
2 P-only DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5/18
2 P-only DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5/18
2 P-only DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11/18
3 ANN10 + P-only DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
3 ANN10 + P-only DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13/18
3 ANN10 + P-only DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
3 ANN10 + P-only DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
3 ANN10 + P-only DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
3 ANN10 + P-only DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5/18
3 ANN10 + P-only DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
4 T/P DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
4 T/P DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8/18
4 T/P DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5/18
4 T/P DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
4 T/P DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12/18
4 T/P DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7/18
5 ANN10 + T/P DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8/18
5 ANN10 + T/P DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7/18
5 ANN10 + T/P DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
5 ANN10 + T/P DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5/18
5 ANN10 + T/P DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
5 ANN10 + T/P DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
5 ANN10 + T/P DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test T/P/ϕ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10/18
6 T/P/ϕ DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
6 T/P/ϕ DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
6 T/P/ϕ DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test T/P/ϕ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4/18
6 T/P/ϕ DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7/18
6 T/P/ϕ DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
6 T/P/ϕ DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Test ANN10 + T/P/ϕ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Test ANN10 + T/P/ϕ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.